

11. Implement a simple top-down parser for context-free grammars using python.

PROGRAM:

```
# Simple Top-Down Parser (Recursive Descent)

def parse_S(string, index):

    # Rule 1: S -> ab

    if index + 1 < len(string):

        if string[index] == 'a' and string[index + 1] == 'b':

            return index + 2

    # Rule 2: S -> aSb

    if index < len(string) and string[index] == 'a':

        next_index = parse_S(string, index + 1)

        if next_index is not None:

            if next_index < len(string) and string[next_index] ==

'b':

                return next_index + 1

    return None

# Main Program

string = input("Enter the string: ")

result = parse_S(string, 0)

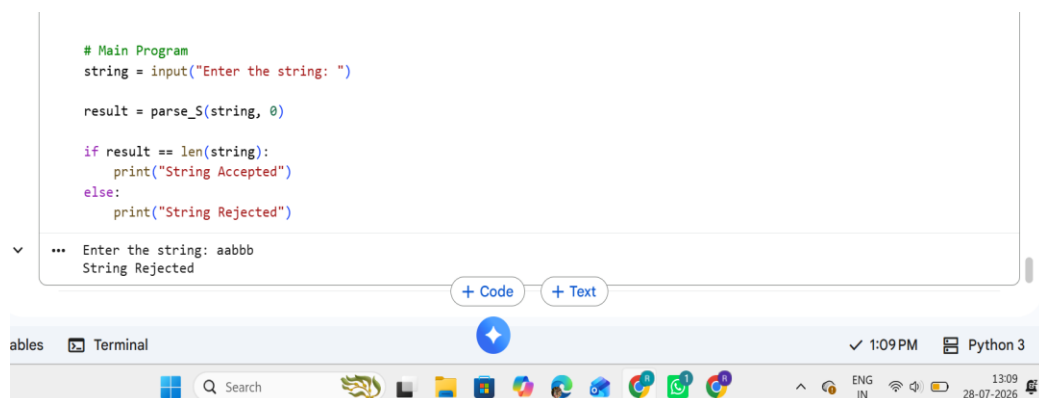
if result == len(string):

    print("String Accepted")

else:

    print("String Rejected")
```

OUTPUT:



```
# Main Program
string = input("Enter the string: ")

result = parse_S(string, 0)

if result == len(string):
    print("String Accepted")
else:
    print("String Rejected")
```

... Enter the string: aabbb
String Rejected

+ Code + Text

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